

Dr. Harender S. Dhattarwal

POSTDOCTORAL ASSOCIATE

Department of Chemistry and Chemical Biology, Rutgers University

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Summary

Highly skilled and experienced computational materials scientist specializing in classical and density functional theory-based molecular dynamics simulations. Expertise in computational materials modeling for next-gen energy storage devices. My research extends to developing new machine learning potentials for simulations with long-range interactions. Skilled in high-performance computing and quantitative data management.

Experience

Department of Chemistry and Chemical Biology, Rutgers University

New Brunswick, NJ, USA

POSTDOCTORAL ASSOCIATE

July 2022 - Present

Advisor: Prof. Richard Remsing

- Developing machine learning potentials for molecular dynamics simulations with long-range interactions.
- Identifying better electrolytes by understanding the hidden electronic disorder in solid-state materials.
- Collaborating with a startup on designing perovskite-based thermochemical energy storage materials.

Department of Chemistry, Indian Institute of Technology Delhi

Delhi, India

DOCTORAL RESEARCH SCHOLAR

July 2017 - July 2022

Advisor: Prof. Hemant K. Kashyap (IIT Delhi, India), Co-Advisor: Prof. Jer-Lai Kuo (IAMS, Taiwan)

- Identified electrolyte intercalation-deintercalation mechanism in nanoscale confinements.
- Studied structure and dynamics of ionic liquids, water-in-salt, and hybrid electrolytes in bulk and near interfaces.
- Wetting-dewetting behavior of carbon surfaces by ionic liquids and deep eutectic solvents (DES).

Institute of Atomic and Molecular Sciences, Academia Sinica

Taipei, Taiwan

SUMMER INTERN

May 2019 - June 2019

- Predicted stability of fluorinated electrolytes for their application in lithium metal batteries.

Institute of Atomic and Molecular Sciences, Academia Sinica

Taipei, Taiwan

WINTER INTERN

Oct. 2018 - Nov. 2018

- Provided mechanistic insights in the solid electrolyte interface formed by superconcentrated electrolyte.

Department of Chemistry, Indian Institute of Technology Guwahati

Guwahati, India

MASTERS RESEARCH SCHOLAR

Dec 2016 - June 2017

Advisor: Prof. Ashish K. Gupta

- Worked on evaluating smooth exterior scale molecular integrals using Monte Carlo method.

Education

Indian Institute of Technology Delhi

Delhi, India

PH.D. IN COMPUTATIONAL CHEMISTRY

2022

Indian Institute of Technology Guwahati

Guwahati, India

M.SC. IN CHEMISTRY

2017

University of Delhi

Delhi, India

B.SC. IN CHEMISTRY

2015

Awards

2024	Best Poster Presentation Award , International Workshop On Advanced Materials (IWAM)	Ras Al-Khaimah, UAE
2021	PCCP Best Poster Presentation Award , Theoretical Chemistry Symposium (TCS)	IISER Kolkata, India
2019	Best Poster Presentation Award , Advanced Simulation Methods: DFT, MD and Beyond (ASM)	IIT Delhi, India
2019	Senior Research Fellowship , University Grants Commission	India
2017	Junior Research Fellowship , University Grants Commission	India

Skills

Programming Python, C++, Fortran, Bash scripting
Scientific LAMMPS, GROMACS, CP2K, VASP, GAUSSIAN, Gauss View
Graphics VMD, Xmgrace, Avogadro, Gnuplot, Matlab, Vector Graphics

Selected Publications (20 Total)

1. **Harender S. Dhatarwal**, Rahul Somni, and Richard C. Remsing, Electronic paddle-wheels in a solid-state electrolyte. *Nat. Commun.* **2024** 15 (121).
2. **Harender S. Dhatarwal** and Hemant K. Kashyap, Unveiling the Influence of Ionic Liquid on the Interfacial Structure and Capacitive Performance of Water-in-Salt Electrolytes at Graphite Electrodes. *J. Phys. Chem. C* **2024** 128 (1), 536.
3. **Harender S. Dhatarwal**, Ang Gao, and Richard C. Remsing, Dielectric Saturation in Water from a Long Range Machine Learning Model. *J. Phys. Chem. B* **2023** 127 (16), 3663.
4. **Harender S. Dhatarwal** and Hemant K. Kashyap, Heterogeneity and Nanostructure of Superconcentrated LiTFSI–EmimTFSI Hybrid Aqueous Electrolytes: Beyond the 21 m Limit of Water-in-Salt Electrolyte. *J. Phys. Chem. B* **2022** 126 (28), 5291.
5. **Harender S. Dhatarwal**, Jer-Lai Kuo, and Hemant K. Kashyap, Mechanistic Insight on the Stability of Ether and Fluorinated Ether Solvent-Based Lithium Bis (fluoromethanesulfonyl) Electrolytes near Li Metal Surface. *J. Phys. Chem. C* **2022** 126 (20), 8953.
6. **Harender S. Dhatarwal** and Hemant K. Kashyap, Unique and Generic Structural Features of Cholinium Amino Acid Biocompatible Ionic Liquids. *Phys. Chem. Chem. Phys.* **2021** 23, 10662.
7. **Harender S. Dhatarwal**, Richard C. Remsing, and Hemant K. Kashyap, Intercalation–deintercalation of water-in-salt electrolytes in nanoscale hydrophobic confinement. *Nanoscale* **2021** 13, 4195.
8. Akshay Malik, **Harender S. Dhatarwal** and Hemant K. Kashyap, Distinct Solvation Structures of CO₂ and SO₂ in Reline and Ethaline Deep Eutectic Solvents Revealed by AIMD Simulations. *J. Phys. Chem. B* **2021** 125, 1852.

Presentations

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| 2024 | Department Seminar , Electronic paddle-wheels accompany diffusion in solid state electrolytes, <i>Department of Metallurgical and Materials Engineering, IIT Ropar</i> | <i>Ropar, India</i> |
| 2024 | Department Seminar , Electronic paddle-wheels accompany diffusion in solid state electrolytes, <i>Department of Physics, IIT Jodhpur</i> | <i>Jodhpur, India</i> |
| 2024 | Poster , Electronic paddle-wheels accompany diffusion in solid state electrolytes, <i>International Workshop on Advanced Materials (IWAM 2024)</i> | <i>Ras Al-Khaimah, UAE</i> |
| 2023 | Poster , Dielectric Saturation in Water Predicted from Self-consistent Field Neural Network (SCFNN) Potential, ACS Fall 2023 | <i>CA, USA.</i> |
| 2023 | Poster , Dielectric Saturation in Water Predicted from Self-consistent Field Neural Network (SCFNN) Potential, <i>Gordon Research Conference on Chemistry and Physics of Liquids (GRC 2023)</i> | <i>CA, USA.</i> |
| 2021 | Talk , Mechanistic Insight on the Solid Electrolyte Interphase (SEI) formed by a Superconcentrated [Li][TFSI] in Acetonitrile Electrolyte near Lithium Metal, <i>The 240th Electrochemical Society Meeting (ECS 2021)</i> | <i>Online</i> |
| 2019 | Poster , Wetting and de-wetting Behavior of Flexible Carbon-based Electrode Materials by Imidazolium-based Ionic Liquid, <i>Advanced Simulation Methods: DFT, MD and Beyond (ASM 2019)</i> | <i>Delhi, India</i> |

Reviewer

- Advanced Materials Interfaces, Wiley, USA.
- Journal of Chemical Theory and Computation, ACS, USA.
- The Journal of Physical Chemistry, ACS, USA.
- ChemPhysChem, Wiley, USA.